

WI3730TU – Numerical Analysis for PDE's

Lecture 8

Numerical time integration:

Forward and Backward Euler methods
convergence, stability
solution of the diffusion equation

Time-dependent problems

Diffusion equation:

$$\frac{\partial u}{\partial t} - \nabla \cdot (k \nabla u) = f$$

Wave equation:

$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} - \Delta u = f$$

Unknown function $u(x, y, t) \Rightarrow$ grid values $u(x_i, y_j, t)$ stored in a vector $\mathbf{u}(t) \Rightarrow$ solution vector changes in time. How to compute the time evolution of $\mathbf{u}(t)$?

Numerical time integration

First-order time evolution

Let $\mathbf{u}(t) \in \mathbb{R}^n$ be a time-evolving vector and $\mathbf{f}(t, \mathbf{u}) \in \mathbb{R}^n$ a vector function depending on both time t and the vector $\mathbf{u}$. Consider a system of the first-order ODE's

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(t, \mathbf{u}), \quad t \geq t_0, \quad \mathbf{u}(t_0) = \mathbf{u}_0$$

$$\frac{d}{dt} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} f_1(t, u_1, u_2, \dots) \\ f_2(t, u_1, u_2, \dots) \\ \vdots \end{bmatrix}$$

For example, a linear system of ODE's

$$\mathbf{u}' = -A\mathbf{u} + \mathbf{s},$$

where A is some matrix. But it can also be a nonlinear function $\mathbf{f}$.

Prediction by time integration

We know $\mathbf{u}(t_0)$ – initial condition. Suppose we want to compute $\mathbf{u}(t_0 + h)$, for some $h > 0$. Integrating the ODE from t_0 to $t_0 + h$, and using the Fundamental Theorem of Calculus (FTC), we get

$$\int_{t_0}^{t_0+h} \mathbf{u}' dt = \int_{t_0}^{t_0+h} \mathbf{f}(t, \mathbf{u}(t)) dt$$

$$\mathbf{u}(t_0 + h) - \mathbf{u}(t_0) = \int_{t_0}^{t_0+h} \mathbf{f}(t, \mathbf{u}(t)) dt$$

$$\mathbf{u}(t_0 + h) = \mathbf{u}(t_0) + \int_{t_0}^{t_0+h} \mathbf{f}(t, \mathbf{u}(t)) dt$$

Numerical approximation

Hence, the problem of finding $\mathbf{u}(t_0 + h)$ reduces to the problem of finding a good numerical approximation (quadrature) for the integral of $\mathbf{f}(t, \mathbf{u})$:

$$\int_{t_0}^{t_0+h} \mathbf{f}(t, \mathbf{u}(t)) dt \approx \sum_{k=1}^K w_k \mathbf{f}(t_k, \mathbf{u}(t_k)), \quad t_0 \leq t_k \leq t_0 + h$$

Having found $\mathbf{u}(t_0 + h)$ we can apply the same procedure to find $\mathbf{u}(t_0 + h + h)$, etc. – time-iterations (recurrence).

Forward-Euler method

Single-point approximation of the integral at the starting point:

$$\int_{t_0}^{t_0+h} \mathbf{f}(t, \mathbf{u}(t)) dt \approx h \mathbf{f}(t_0, \mathbf{u}(t_0))$$

Explicit one-step prediction:

$$\mathbf{u}(t_0 + h) \approx \mathbf{u}(t_0) + h \mathbf{f}(t_0, \mathbf{u}(t_0))$$

Forward-Euler (FE) iterations:

$$\mathbf{u}(t_k + h) \approx \mathbf{u}(t_k) + h \mathbf{f}(t_k, \mathbf{u}(t_k)), \quad k = 0, 1, \dots$$

Backward-Euler method

Single-point approximation of the integral at the end point:

$$\int_{t_k}^{t_{k+1}} \mathbf{f}(t, \mathbf{u}(t)) dt \approx h \mathbf{f}(t_{k+1}, \mathbf{u}(t_{k+1}))$$

Implicit BE iterations

$$\mathbf{u}(t_{k+1}) \approx \mathbf{u}(t_k) + h \mathbf{f}(t_{k+1}, \mathbf{u}(t_{k+1})), \quad k = 0, 1, \dots$$

- This requires solving a vector equation to find $\mathbf{u}(t_{k+1})$. The Backward-Euler is an *implicit* method, whereas, the Forward-Euler is an *explicit* method

Trapezoidal method

Two-point approximation of the integral:

$$\int_{t_k}^{t_{k+1}} \mathbf{f}(t, \mathbf{u}(t)) dt \approx \frac{1}{2}h[\mathbf{f}(t_k, \mathbf{u}(t_k)) + \mathbf{f}(t_{k+1}, \mathbf{u}(t_{k+1}))]$$

Implicit trapezoidal iterations

$$\mathbf{u}(t_{k+1}) \approx \mathbf{u}(t_k) + \frac{1}{2}h[\mathbf{f}(t_k, \mathbf{u}(t_k)) + \mathbf{f}(t_{k+1}, \mathbf{u}(t_{k+1}))], \quad k = 0, 1, \dots$$

Other integration methods

$$\int_{t_0}^{t_0+h} \mathbf{f}(t, \mathbf{u}(t)) dt \approx \sum_{i=1}^k w_i \mathbf{f}(t_k, \mathbf{u}(t_k)), \quad t_0 \leq t_k \leq t_0 + h$$

- Multistep methods: Adams-Bashforth, backward differentiation formulae (BDF)
- Explicit and implicit Runge-Kutta methods (ERK, IRK)

Convergence and order of time integration methods

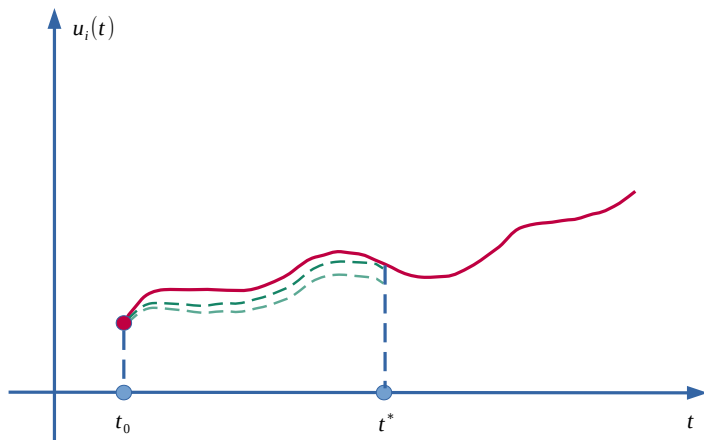
Convergence of time integration methods

- Suppose we have applied some time integration method on the interval $t \in [t_0, t^*]$ with the time step $h = (t^* - t_0)/K$, and obtained the numerical approximation $\mathbf{u}_h(t_k)$ of the actual $\mathbf{u}(t_k)$, for all $k = 0, 1, \dots, K$.
- We say that the method is *convergent* if for every $\mathbf{f}(t, \mathbf{u})$ and every $t^* > t_0$ it is true that

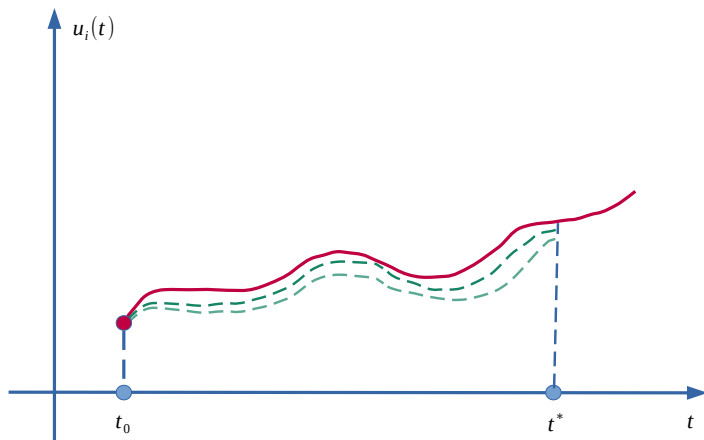
$$\lim_{h \rightarrow 0^+} \max_{k=0,1,\dots,K} \|\mathbf{u}_h(t_k) - \mathbf{u}(t_k)\|_2 = 0$$

Global max-2-norm converges to zero as time step h decreases for any finite t^* .

Idea of convergence



Idea of convergence



Convergence and global error of the Forward-Euler method

Theorem: Let $\mathbf{u}(t)$ be continuously differentiable and $\mathbf{f}(t, \mathbf{u})$ a *Lipschitz* function, i.e., there is $\lambda > 0$ such that $\|\mathbf{f}(t, \mathbf{u}) - \mathbf{f}(t, \mathbf{v})\| \leq \lambda \|\mathbf{u} - \mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Then, the Forward-Euler method is convergent

Proof: Denoting $\mathbf{e}_h(t_k) = \mathbf{u}_h(t_k) - \mathbf{u}(t_k)$

$$\mathbf{u}(t_{k+1}) = \mathbf{u}(t_k) + h\mathbf{u}'(t_k) + \mathcal{O}(h^2) = \mathbf{u}(t_k) + h\mathbf{f}(t_k, \mathbf{u}(t_k)) + \mathcal{O}(h^2)$$

$$\mathbf{e}_h(t_{k+1}) = \mathbf{e}_h(t_k) + h[\mathbf{f}(t_k, \mathbf{u}(t_k) + \mathbf{e}_h(t_k)) - \mathbf{f}(t_k, \mathbf{u}(t_k))] + \mathcal{O}(h^2)$$

$$\|\mathbf{e}_h(t_{k+1})\| \leq \|\mathbf{e}_h(t_k)\| + h\|\mathbf{f}(t_k, \mathbf{u}(t_k) + \mathbf{e}_h(t_k)) - \mathbf{f}(t_k, \mathbf{u}(t_k))\| + Ch^2$$

$$\|\mathbf{e}_h(t_{k+1})\| \leq (1 + h\lambda)\|\mathbf{e}_h(t_k)\| + Ch^2, \quad k = 0, 1, \dots, n$$

$$\|\mathbf{e}_h(t_k)\| \leq \frac{C}{\lambda} h [(1 + h\lambda)^k - 1] < \frac{C}{\lambda} h (e^{kh\lambda} - 1) \leq \frac{C}{\lambda} (e^{\lambda t^*} - 1) h$$

$$\lim_{h \rightarrow 0^+} \max_k \|\mathbf{e}_h(t_k)\| \leq \frac{C}{\lambda} (e^{\lambda t^*} - 1) \lim_{h \rightarrow 0^+} h = 0$$

Local and global orders of time integration methods

- We say that the time integration method is *locally of order p* if

$$\mathbf{u}_h(t_1) - \mathbf{u}(t_1) = \mathcal{O}(h^p)$$

- Method of local order p recovers exactly any polynomial of degree $p - 1$ or less
- Forward Euler method is locally $\mathcal{O}(h^2)$. Indeed:

$$\begin{aligned}\mathbf{u}_h(t_1) - \mathbf{u}(t_1) &= \mathbf{u}_h(t_0) + h\mathbf{f}(t_0, \mathbf{u}_h(t_0)) \\ &\quad - \mathbf{u}(t_0) - h\mathbf{f}(t_0, \mathbf{u}(t_0)) = \mathcal{O}(h^2) = \mathcal{O}(h^2),\end{aligned}$$

since $\mathbf{u}_h(t_0) = \mathbf{u}(t_0)$. *Global* error drops to order $p - 1$, and convergence analysis (previous slide) shows that the global error is $\|\mathbf{e}_h(t_k)\| = \mathcal{O}(h)$

Convergence and order of other methods

- Backward Euler method is convergent and locally $\mathcal{O}(h^2)$, globally $\mathcal{O}(h)$.
- Trapezoidal rule is an implicit, convergent, locally $\mathcal{O}(h^3)$ and globally $\mathcal{O}(h^2)$ method

$$\|\mathbf{e}_h(t_k)\| \leq \frac{Ch^2}{\lambda} \exp\left(\frac{\lambda t^*}{1 - \lambda h/2}\right), \quad k = 0, 1, \dots, K$$

Method of lines for diffusion equation

Method of lines discretizes in space, but keeps time continuous. Serves as a theoretical benchmark in stability analysis of numerical time-integration methods.

With FDM/FVM spatial discretization

Later we shall see that after FDM discretization in space:

$-\Delta u(\mathbf{x}, t) \rightarrow A\mathbf{u}(t)$, $\mathbf{u}(t) \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$, the diffusion equation becomes the system of ODE's:

$$\mathbf{u}' = -A\mathbf{u}, \quad \mathbf{u}(0) = \mathbf{u}_0$$

Assume $A = V\Lambda V^{-1}$ exists. Then, $\mathbf{u}(t) = V\mathbf{b}(t)$, and $\mathbf{b}(t) = V^{-1}\mathbf{u}(t)$.

$$V\mathbf{b}' = -V\Lambda V^{-1}\mathbf{u}, \quad \mathbf{u}_0 = V\mathbf{b}_0$$

$$\mathbf{b}' = -\Lambda\mathbf{b}, \quad \mathbf{b}(0) = V^{-1}\mathbf{u}_0,$$

$$b_j'(t) = -\lambda_j b_j(t), \quad j = 1, \dots, n$$

$$b_j(t) = b_j(0)e^{-\lambda_j t}, \quad j = 1, \dots, n$$

$$\mathbf{b}(t) = e^{-\Lambda t}\mathbf{b}(0) = e^{-\Lambda t}V^{-1}\mathbf{u}(0)$$

$$\mathbf{u}(t) = Ve^{-\Lambda t}V^{-1}\mathbf{u}(0)$$

This is the exact solution of the discrete time-evolution problem.

Case of (negative) FD Laplacian

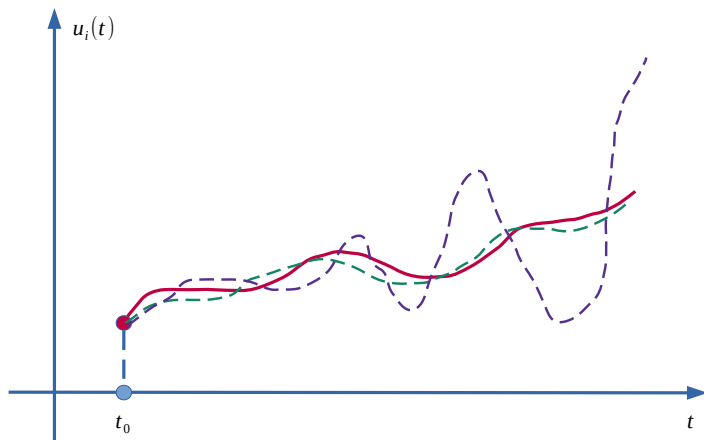
For $-\Delta \rightarrow A$ with the FDM method the eigenvalues are positive $\lambda_j > 0$, $j = 1, \dots, n$. Writing out $\mathbf{u}(t) = V\mathbf{b}(t)$ and using the method-of-lines to find $\mathbf{b}(t)$, we get

$$\mathbf{u}(t) = \sum_{j=1}^n b_j(t) \mathbf{v}_j$$
$$b_j(t) = b_j(0) e^{-\lambda_j t}$$

Shows that the amplitudes $b_j(t)$ of all expansion terms $b_j(t) \mathbf{v}$ of the solution decay exponentially in time.

Stability of time integration methods

Idea of stability



Stability of the forward Euler method

After FDM/FVM discretization the diffusion equation becomes

$$\mathbf{u}' = -A\mathbf{u}, \quad \mathbf{u}(t_0) = \mathbf{u}_0$$

the Forward-Euler method with the time step h produces iterations $k = 0, 1, \dots$:

$$\mathbf{u}_h(t_k) = \mathbf{u}_h(t_{k-1}) - hA\mathbf{u}_h(t_{k-1}) = [I - hA]\mathbf{u}_h(t_{k-1}) = [I - hA]^k \mathbf{u}_0$$

Let A be diagonalizable: $A = V\Lambda V^{-1}$ with eigenvalues λ_j , $j = 1, \dots, n$.

$$\mathbf{u}(t_k) = V e^{\Lambda t_k} V^{-1} \mathbf{u}_0 = \sum_{j=1}^n b_j(0) e^{-\lambda_j t_k} \mathbf{v}_j \quad (\text{method of lines})$$

$$\mathbf{u}_h(t_k) = [I - hA]^k \mathbf{u}_0 = V [I - h\Lambda]^k V^{-1} \mathbf{u}_0 = \sum_{j=1}^n b_j(0) (1 - h\lambda_j)^k \mathbf{v}_j$$

Hence, for $\lambda_j > 0$, $e^{-\lambda_j t_k}$ always decays with t_k , but $|1 - h\lambda_j|^k$ may be growing with k if h is too large. The Forward-Euler method is not always stable.

Conditional stability of the forward Euler method

Deriving the condition for the stability of the Forward-Euler method:

$$\begin{aligned} |1 - h\lambda_j| &< 1 \\ -1 &< 1 - h\lambda_j < 1 \\ -2 &< -h\lambda_j < 0 \\ 0 &< h\lambda_j < 2 \\ 0 &< h < \frac{2}{\lambda_j}, \quad j = 1, \dots, n \end{aligned}$$

It is common to denote the time step as $h = \Delta t$:

$$0 < \Delta t < \frac{2}{\max_j \lambda_j}$$

Unconditional stability of Backward-Euler method

Backward-Euler method with $\Delta t = h$ produces iterations $k = 0, 1, \dots$:

$$\mathbf{u}_h(t_k) = \mathbf{u}_h(t_{k-1}) - hA\mathbf{u}_h(t_k)$$

$$[I + hA]\mathbf{u}_h(t_k) = \mathbf{u}(t_{k-1})$$

$$\mathbf{u}_h(t_k) = [I + hA]^{-1}\mathbf{u}_{t_{k-1}} = [I + hA]^{-k}\mathbf{u}_0$$

$$\mathbf{u}_h(t_k) = [I + hA]^{-k}\mathbf{u}_0 = V[I + h\Lambda]^{-k}V^{-1}\mathbf{u}_0 = \sum_{j=1}^n b_j(0) \frac{1}{(1 + h\lambda_j)^k} \mathbf{v}_j$$

Hence, for $\lambda_j > 0$, the Backward-Euler method is stable for any $h > 0$

Unconditional stability of trapezoidal method

Trapezoidal method with $\Delta t = h$ produces iterations $k = 0, 1, \dots$:

$$\mathbf{u}_h(t_k) = \mathbf{u}_h(t_{k-1}) - \frac{1}{2}hA[\mathbf{u}_h(t_{k-1}) + \mathbf{u}_h(t_k)]$$

$$\left[I + \frac{h}{2}A \right] \mathbf{u}_h(t_k) = \left[I - \frac{h}{2}A \right] \mathbf{u}(t_{k-1})$$

$$\mathbf{u}_h(t_k) = \left[I + \frac{h}{2}A \right]^{-1} \left[I - \frac{h}{2}A \right] \mathbf{u}_{t_{k-1}} = \left[I - \frac{h}{2}A \right]^k \left[I + \frac{h}{2}A \right]^{-k} \mathbf{u}_0$$

$$\mathbf{u}_h(t_k) = \sum_{j=1}^n b_j(0) \left(\frac{1 - \lambda_j h/2}{1 + \lambda_j h/2} \right)^k \mathbf{v}_j$$

With $\lambda_j > 0$ trapezoidal method is stable for any $h > 0$

Solving the diffusion equation

Diffusion equation in 2D

$$\frac{\partial u}{\partial t} - \Delta u = f, \quad (x, y) \in \Omega, \quad 0 < t \leq T$$

$$u(x, y, t) = g(x, y, t), \quad (x, y) \in \partial\Omega, \quad 0 < t \leq T$$

$$u(x, y, 0) = u_0(x, y), \quad (x, y) \in \Omega, \quad t = 0$$

Sometimes it is written as $(x, y, t) \in \Omega \times (0, T]$

Time integration: Forward-Euler method

Although $u(x, y, t)$ is a function rather than a vector, all previously explained time integration methods can still be applied. With the time step h the Forward-Euler time iterations are

$$\begin{aligned}\frac{\partial u(x, y, t)}{\partial t} &= \Delta u(x, y, t) + f(x, y, t) \\ u(x, y, t_{k+1}) &= u(x, y, t_k) + h [\Delta u(x, y, t_k) + f(x, y, t_k)] \\ u^{k+1}(x, y) &= u^k(x, y) + h [\Delta u^k(x, y) + f^k(x, y)] \\ k &= 0, 1, \dots; \quad u(x, y, t_0) = u_0(x, y)\end{aligned}$$

The time-dependent boundary condition $u(x, y, t_k) = g(x, y, t_k)$, $(x, y) \in \partial\Omega$ must also be enforced at each t_k .

Time integration: Backward-Euler method

The Backward-Euler time iterations are

$$\frac{\partial u(x, y, t)}{\partial t} = \Delta u(x, y, t) + f(x, y, t)$$

$$u(x, y, t_{k+1}) = u(x, y, t_k) + h [\Delta u(x, y, t_{k+1}) + f(x, y, t_{k+1})]$$

$$u^{k+1}(x, y) - h \Delta u^{k+1}(x, y) = u^k(x, y) + h f^{k+1}(x, y)$$

$$k = 0, 1, \dots; \quad u(x, y, t_0) = u_0(x, y)$$

At the $(k + 1)$ -st iteration:

- $u^k(x, y)$ and $f^{k+1}(x, y)$ are known functions of $(x, y) \in \Omega$
- $u^{k+1}(x, y)$ is an unknown function

Space discretization: FDM or FVM

- In both FDM and FVM, introduce spatial grid and the solution vector $\mathbf{u}^k \in \mathbb{R}^n$, related to the grid values $u(x_i, y_j, t_k)$. Discretize the Laplacian operator $-\Delta \rightarrow A \in \mathbb{R}^{n \times n}$, moving Dirichlet boundary conditions to the right-hand side $\mathbf{f}^k$ as usual
- For example, Backward-Euler iteration becomes

$$u^{k+1}(x, y) - h\Delta u^{k+1}(x, y) = u^k(x, y) + hf^{k+1}(x, y)$$

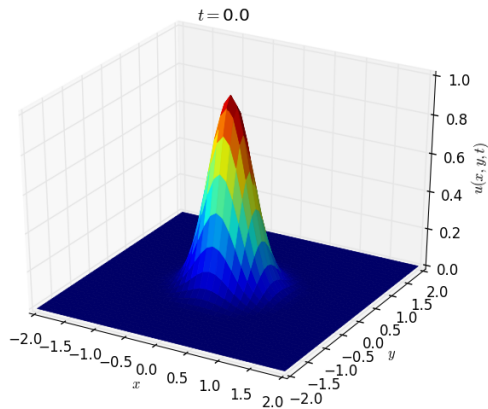
$$\mathbf{u}^{k+1} + hA\mathbf{u}^{k+1} = \mathbf{u}^k + h\mathbf{f}^{k+1}$$

$$[I + hA] \mathbf{u}^{k+1} = \mathbf{u}^k + h\mathbf{f}^{k+1}$$

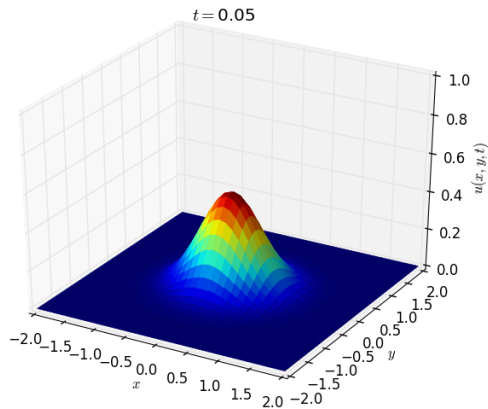
$$k = 0, 1, \dots; \quad \mathbf{u}^0 = \mathbf{u}_0$$

- To find the current grid values $\mathbf{u}^{k+1}$ in backward Euler we need to solve a linear system. This is computationally equivalent to solving one steady-state problem at each iteration

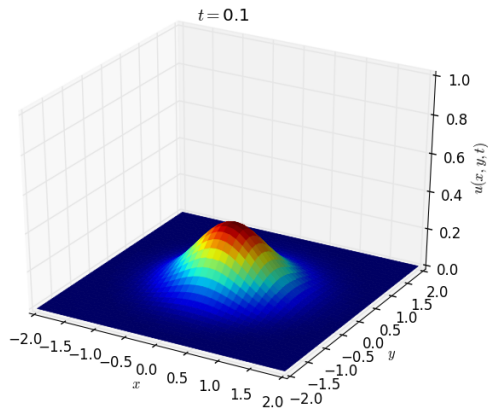
Diffusion: $t = 0.0$ (initial condition)



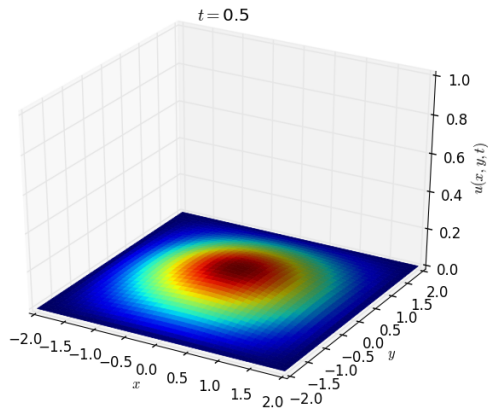
Diffusion: $t = 0.05$



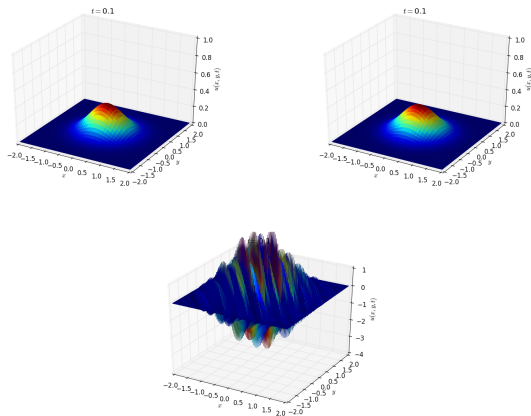
Diffusion: $t = 0.1$



Diffusion: $t = 0.5$



Backward vs Forward Euler: $t = 0.1$



Left: backward Euler with $\Delta t = 0.01$ (10 steps)

Middle and right: forward Euler with $\Delta t = 0.0007$ (150 steps) and
 $\Delta t = 0.00095$ (105 steps)